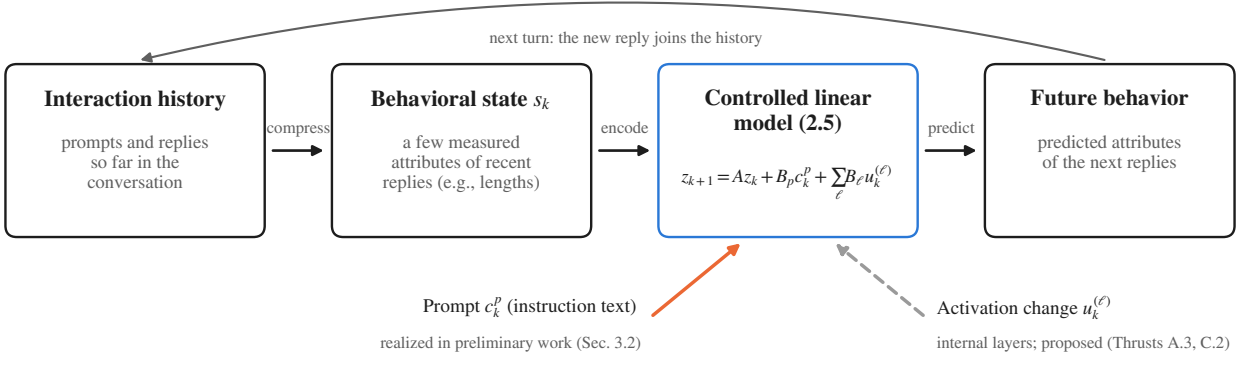
next turn: the new reply joins the history
Interaction history
prompts and replies so far in the conversation
compress
Behavioral state s_k
a few measured attributes of recent replies (e.g., lengths)
encode
Controlled linear model (2.5)
$z_{k+1} = Az_k + B_p c_k^p + \sum_\ell B_\ell u_k^{(\ell)}$
predict
Future behavior
predicted attributes of the next replies
Prompt c_k^p (instruction text)
realized in preliminary work (Sec. 3.2)
Activation change $u_k^{(\ell)}$
internal layers; proposed (Thrusts A.3, C.2)